

CIA triads in Cyber security

The **CIA Triad** is the foundation of **Cyber Security**. It represents the three main goals of information security:

C – confidentiality

I – integrity

A – Availability

1. Confidentiality (C)

What is Confidentiality?

Confidentiality means protecting information from unauthorised access or disclosure. Only the people who have permission should be able to see or use the data.

Simple :- confidentiality means information or data only access for the right person

Example:- ATM PIN is confidential.

- You know your PIN.
- The bank system knows your PIN.
- Other people should not know your PIN.

2. Integrity(I)

What is Integrity?

Integrity means maintaining the accuracy, correctness, and trustworthiness of data. Data should not be modified, deleted, or changed without permission.

Simple:- Integrity means a information or data change without the permission

Example:- you share a message to your friend say “HI”

Original:- HI the attacker change :- Bye integrity is broken

3. Availability

What is Availability?

Availability means ensuring that information and systems are accessible whenever authorized users need them.

Simple:- availability means a data is always work we need to them

Example:- you cannot take bank statement by server downAvailability is affected

Real-World Example of CIA Triad

Online Banking System

Confidentiality

Login credentials and account information remain private.

Integrity

Transactions cannot be altered without authorization.

Availability

Customers can access banking services whenever needed.

Security Objectives of the CIA Triad

The CIA Triad supports important security objectives:

1. **Protect Information** Secure sensitive and valuable data.
2. **Reduce Security Risks** Lower the chance of cyber incidents.
3. **Maintain Trust** Build confidence among users and organizations.
4. **Support Business Continuity** Ensure services remain operational.
5. **Improve Reliability** Maintain consistent system performance.
6. **Support Compliance** Help organizations follow security standards and regulations.